Requirement Specification: Smart City Traffic Management

Group 2 · Reasoning and Decision Making under Uncertainty (MAIB DSC 103) · Version 2.1, 12 November 2026 (replaces v2.0 of 8 November)
Live dashboard (live simulation, no installation): https://smarttrafficu-mmzjikh5bg8frby3adzdc2.streamlit.app/ · Colab code (open to anyone with the link): colab.research.google.com/github/Sbot-c/Smart_Traffic_U/.../Smart_Traffic_MARL_DecisionNetwork.ipynb · Code and data: github.com/Sbot-c/Smart_Traffic_U

Dashboard view	Owner	Responsible for
1 City Overview	Sparsh	multi-agent system on the map: live-simulation replay, KPIs vs fixed-time
2 RL Simulation	Shashaank	Q-learning and SARSA agents: decisions, learned policies, training
3 Decision Network Visualizer	Siddal	influence diagram, expected utility, value of information
4 Scenario Explorer	Aradhana	seven scenarios, parameters and outcome distributions
5 KPI & Trade-off Analysis	Shirley	improvement heatmaps, delay vs safety, course techniques
6 Policy Comparison	Sparsh	signals vs routing, tuned baseline, paired statistics
7 Simulation Lab	Aradhana	live what-if with linked conditions and sensing bias
8 Research Extensions	Shashaank	AV speed and lanes, bus loads, safety guard, sensing uncertainty

Team table written by the group. Every section inside an amber dashed frame was drafted with GenAI and then checked and edited by the group; the prompts are in GenAI_disclosure_and_prompts.md.

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1. Problem statement, objectives and requirements

Problem. A city traffic authority must choose how to control a signalised network when demand, weather, pedestrians, incidents and emergencies are uncertain, and must trade off efficiency, safety and emissions. We specify a decision-support system for a 3 × 3 Downtown-Dubai-style grid (nine junctions, 400 m blocks, a three-lane eastbound corridor on Street 2) in which traffic lights, autonomous vehicles (AVs) and buses are learning agents (Q-learning and SARSA), a decision network chooses the control strategy, and a dashboard shows the evidence as a live simulation of simulated data. It is a research and teaching prototype on synthetic demand, not a deployed controller. The starting point is Wairagade, Ugale & Waghmare (2025), who propose RL signal timing with an emergency override and a monitoring dashboard but report no quantitative evaluation; this project builds and evaluates such a system.

Stakeholders. Road-authority operators (choose the strategy), commuters and AV operators (travel time), bus operators and passengers (punctuality, crowding), pedestrians (waiting, safety), emergency services (pre-emption), residents (emissions).

ID	Objective	Measured by (KPI)
O1	Reduce congestion and delay	average delay (s/veh), throughput (veh/h), stops
O2	Improve safety for vehicles and pedestrians	expected conflicts per 1,000 veh, jaywalks, pedestrian wait
O3	Lower emissions	CO_2 (kg), NOx (g), idle time; health-impact node
O4	Keep public transport reliable	bus on-time %, bus delay, load, left behind
O5	Decide well under uncertainty	expected utility, value of information, weight and sensing sensitivity

Functional requirements

ID	The system shall …	Implemented by
FR1	simulate a signalised grid with time-varying demand, turns, queues, spill-back and lost time	traffic_env.py: 5 s steps, 30-min episodes, 5-min warm-up
FR2	model peak demand, pedestrian crossings and jaywalking, emergency-vehicle pre-emption, rain, sandstorm, incidents and an arena event	scenario parameters (below); pedestrians and emergency vehicles run in every scenario
FR3	let traffic-light agents keep or change phase	Q-learning / SARSA, one table per junction (2,250 states)
FR4	let AV agents route cooperatively, with speed control and lane selection	RL routing (64 states); V2I speed advisory; lane selection
FR5	let bus agents request priority or hold; measure loads	RL bus agents (144 states): go / priority / hold 20 s; load KPIs
FR6	keep every controller inside a safety shield	min/max green, pedestrian clearance, EV pre-emption, bus priority; optional pedestrian max-wait guard
FR7	compare against fixed-time and actuated baselines	Webster plan; actuated rule; actuated tuned on validation runs; signal × routing ablation
FR8	evaluate strategies with a decision network	chance, decision and utility nodes; expected utility; value of information (section 3.5)
FR9	report every KPI with its uncertainty	8 held-out test episodes, mean ± s.d., paired tests; robustness studies
FR10	provide an interactive dashboard and export the data	Streamlit, 8 views (section 5); CSV download

Non-functional requirements

ID	Requirement	Target / evidence
NFR1	Reproducibility	fixed seeds (training; validation 800,000+; test 900,000+); common random numbers; every CSV regenerated by scripts
NFR2	Honest evaluation	choices made on validation runs only; test episodes never used to choose; spread over three re-trainings reported
NFR3	Performance	≈0.06-0.17 s per 30-min episode on a CPU; full pipeline in minutes; the Lab re-runs four policies in about a second
NFR4	Explainability	tabular Q-values, logged decisions and shield overrides, readable probabilities
NFR5	Usability and access	hosted link, no installation; plain-language captions; best viewed full screen; tiles wrap in narrow windows

Scenarios (all 30 simulated minutes; pedestrians and emergency vehicles in each)

Scenario	Key parameters
Normal flow	demand 1.0, clear, bus every 6 min, 3 emergency calls / 30 min
Rush hour	demand 1.3 ramping to ≈1.5, eastbound corridor +15 %, pedestrians 1.3, bus passengers 1.8
Incident	crash before J5 eastbound, minutes 5-25: capacity −75 %, speed −60 %, 6 emergency calls
Rain / Sandstorm	capacity −6 % / −12 %, speed −10 % / −20 %, crash risk ×1.71 / ×2.0, demand ×0.95 / ×0.85, pedestrians ×0.60 / ×0.30, 4 emergency calls
Event surge	arena lets out at minute 8: cars and pedestrians surge at three junctions, bus passengers ×1.5
Bus corridor	dedicated bus lane on Street 2, bus every 3 min, passengers ×2.0

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2. Data and data sources

All traffic data are **generated by the simulator** (there is no public, junction-level Dubai feed for these conditions); the parameters that drive it come from the published sources below, and the values marked *assumption* are our modelling choices (listed in *config.py*).

Data file (data/)	Rows	Content
smart_traffic_simulation_log.csv (main)	15,120	one row per junction per 30 s window, every scenario × policy: queues, arrivals, departures, delay, pedestrians, jaywalks, conflicts, CO_2, NOx, stops, spill-back, EV and bus flags, reward
kpi_results.csv	224	KPIs per test episode (7 scenarios × 4 policies × 8)
decision_network_samples.csv	11,664	junction × 10-min samples used to learn the decision-network probabilities
learning_curves.csv · model_selection.csv	1,680 · 14	training metrics; general vs adapted tables on validation runs
ablation_results.csv · training_robustness.csv	400 · 18	signal × routing and component studies; re-training with other seeds
ext_*.csv (incl. ext_sensing*.csv)	—	research extensions and the sensing-uncertainty study (2,072 episodes)
q_tables.npz · animations.json.gz	—	trained Q-tables; replay trajectories

Parameter	Value used	Source
Network layout	3 × 3 grid, ≈400 m blocks	Downtown Dubai / Business Bay street grid (approximation)
Saturation flow, lost time, pedestrian non-compliance	1,800 veh/h/lane; patience 30 s	Transportation Research Board (2016), Highway Capacity Manual, 6th ed.
Fixed-time plan	Webster optimum cycle	Webster (1958), Road Research Technical Paper 39
Rain: capacity, speed	−6 %, −10 %	Perrin et al. (2002) in Sullivan, Sisiopiku & Reddy-Surhabi (2010), UTCA 07211
Rain: crash rate	+71 %	Qiu & Nixon (2008), Transportation Research Record 2055
Sandstorm (low visibility): capacity	−12 %	HCM via FHWA (2013), FHWA-HOP-13-042; speed −20 % and crash ×2.0 are assumptions
CO_2 and idling	400 g CO_2/mile; 0.3 / 1 gal/h idle	US EPA (2023); Argonne National Laboratory idling fact sheet
Value of time	USD 14.10 /person-h = AED 51.8	U.S. DOT value of travel time (US figure, editable)
AV share, platooning, red-light running	25 %; 0.85 headway; 8 %	assumptions

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3. Algorithms and methods

3.1 Agents (each with its own state, actions and reward)

Agent	State (discretised)	Actions	Reward (maximised = minus cost)
Traffic light ×9	phase, green time, queue on green and red, arrivals on green, pedestrian / spill-back flag (2,250)	keep / change phase every 5 s	−(delay + 0.5 × pedestrian delay + 300 × expected conflicts + 0.2 × g CO_2) / 100
Autonomous vehicle (fleet)	remaining moves, congestion of the two candidate links (64)	next link; V2I speed advisory; lane selection	−(travel time + congestion imposed on others)
Bus (fleet)	schedule deviation, load, headway, signal ahead, cross queue (144)	go / request priority / hold 20 s	−(passenger-minutes + schedule deviation + priority cost to cross traffic)

3.2 Learning rules (tabular, ε-greedy)

Q-learning (off-policy): $Q(s,a) \leftarrow Q(s,a) + \alpha\,[\, r + \gamma \max_{a'} Q(s',a') - Q(s,a)\,]$

SARSA (on-policy): $Q(s,a) \leftarrow Q(s,a) + \alpha\,[\, r + \gamma\, Q(s',a') - Q(s,a)\,]$, with a' the action actually taken

$\alpha = 0.10$, $\gamma = 0.90$ (AV routing 0.98), ε from 0.30 to 0.02 over the first 80 % of episodes. Q starts at 0; in a state never visited in training the signal falls back to the actuated rule (at most 7 of 25,920 decisions per scenario). Every action passes the safety shield (FR6). Tables, not deep RL: 2,250 states per junction, every decision can be inspected, training takes minutes on a CPU.

3.3 Training and evaluation protocol

280 pre-training episodes over all scenarios (the nine junctions share one table; weather, pedestrian level and demand are randomised so the policy is robust to them), then **80 adaptation episodes per scenario** with nine independent tables. For each scenario the general or the adapted table is chosen on 8 validation episodes; results are reported on 8 separate test episodes with the same arrivals for every policy. Comparisons use paired t-tests. Baselines: Webster fixed-time; vehicle-actuated (gap-out, queue ratio, pedestrian service, 60 s max green); the actuated rule tuned per scenario on validation runs (81 settings).

3.4 Policy packages

Actuated = actuated signals + static routes; *Q-learning / SARSA* = RL signals + cooperative AV routing. Signal controllers are compared with the same routing (signal × routing ablation), so routing gains are not credited to the signals.

3.5 Decision network (influence diagram)

Chance nodes Weather {clear, rain, sandstorm}, Traffic density and Pedestrian flow {low, medium, high} → Congestion, Collision risk, Emissions → Health impact; decision nodes Signal timing {fixed, actuated, Q-learning, SARSA} × Routing {static, cooperative} (8 options); utility nodes efficiency, safety, environment with weights 0.40 / 0.35 / 0.25 (adjustable). Conditional probabilities are learned from the 11,664 simulated samples with a Dirichlet(1) prior (Laplace smoothing); expected utility is computed by enumeration (cross-checked with pgmpy variable elimination). Outputs: the best decision per condition, value of information (EVPI) and weight sensitivity. Result: actuated signals + cooperative routing in 6 of 7 scenarios (rain: + static routes), i.e. in 22 of 27 weather × density × pedestrian conditions; these two counts measure different things.

3.6 Where the uncertainty is, and how it is handled

(a) **Sensing / likelihood bias.** Real queue counts come from detectors or phone data; counting phones and assuming one per person over-counts when people carry two, low coverage under-counts. The deployed agents were re-run with counts biased 0.5×-2× and with Poisson noise (2,072 test episodes, *sensing.py*). Within 0.75-1.5× the delay changes by +1.4 % (actuated), +3.2 % (Q-learning) and +3.2 % (SARSA); worst case event surge at 2×: Q-learning 46.6 → 60.2 s, still ahead of actuated (65.5 s). The RL-vs-actuated verdict is unchanged in 86 % of 84 conditions (Figure 3). (b) **Value of information** ≤ 0.004 utility points: observing a condition first would not change the decision. (c) **Safety model**: halving or doubling each conflict mechanism keeps the safest policy in 90 % of 42 cases. (d) **Training randomness**: Q-learning re-trained three times gives 72.5 ± 2.1 s in rush hour. (e) The network's probabilities are smoothed estimates learned from simulation, not measured frequencies.

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4. Required libraries and tools

Library / tool	Version	Used for
Python	3.11 or later	everything
numpy	≥ 1.26	simulator state, Q-tables, random streams
pandas	≥ 2.0	logs, KPIs, CSV export, decision-network samples
scipy	≥ 1.11	paired t-tests and statistics
streamlit	≥ 1.49	dashboard (app.py); hosted on Streamlit Community Cloud
plotly	≥ 5.20	dashboard charts and network map
matplotlib	any recent	notebook and report figures
pgmpy	any recent (installed in the notebook)	cross-check of the decision-network inference
Google Colab · GitHub · Streamlit Community Cloud	—	notebook (code link), source and data, public dashboard

Install: pip install -r requirements.txt. Hardware: an ordinary CPU; no GPU.

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5. Dashboard design (user-interface wireframe)

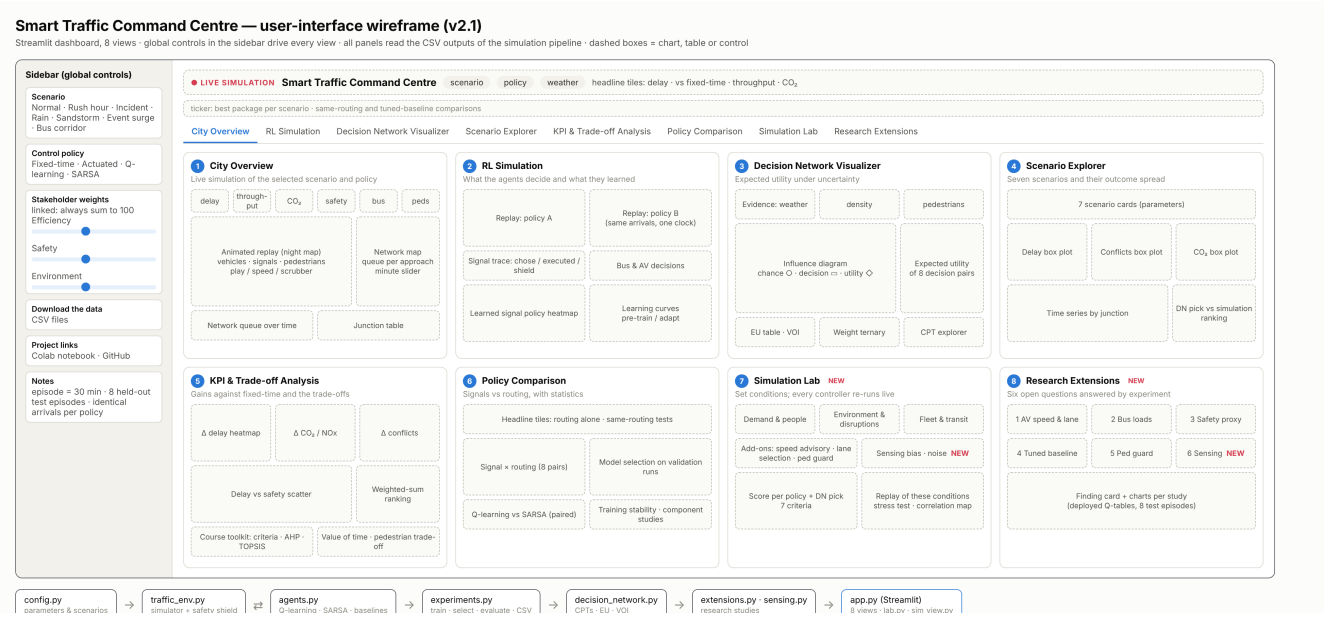


Figure 1. Wireframe of the eight views. The sidebar (scenario, policy, linked stakeholder weights, downloads, links) drives every view; the header is labelled live simulation because it replays and re-runs simulated data.

View	Must show
City Overview	animated replay with signals, queues and counters; KPI tiles vs fixed-time; network map with a minute slider
RL Simulation	side-by-side replay of two policies on the same arrivals; agent decisions; learned policy; learning curves
Decision Network Visualizer	influence diagram with evidence; expected utility of all 8 options; value of information; weight sensitivity; probability tables
Scenario Explorer	the seven scenarios, their parameters and KPI distributions
KPI & Trade-off Analysis	improvement heatmaps; delay vs safety; course techniques (back-up material)
Policy Comparison	signal × routing ablation with paired statistics; model selection; Q-learning vs SARSA
Simulation Lab	live re-run of all policies under linked conditions, add-ons and sensing bias; replay; stress test
Research Extensions	six studies: AV speed and lanes, bus loads, safety proxy, tuned baseline, pedestrian guard, sensing uncertainty

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Appendix A. Results and acceptance (8 held-out test episodes per result)

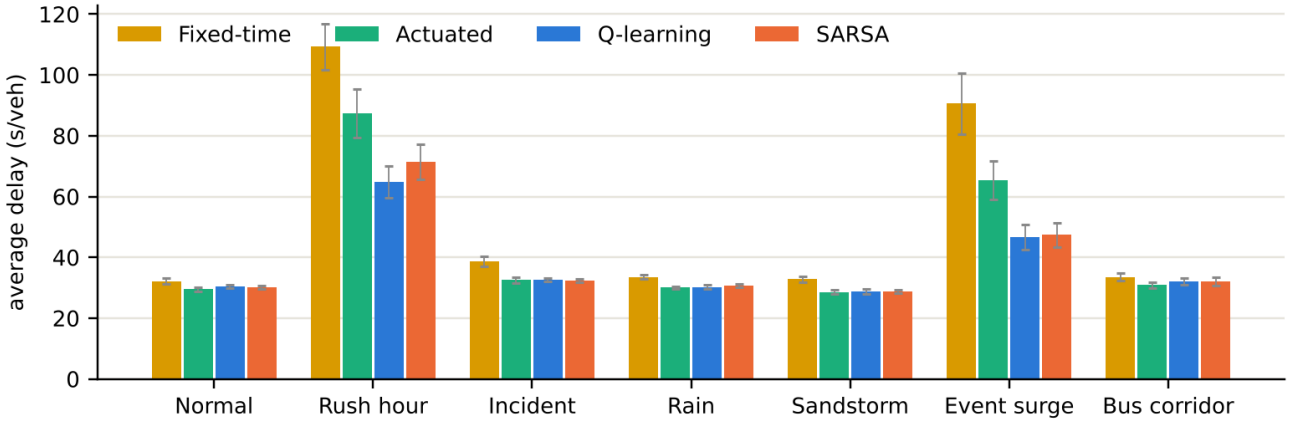


Figure 2. Average delay by scenario and policy package (mean ± s.d.).

Criterion	Result
Delay vs fixed-time (packages)	Q-learning package −41 % in rush hour (64.8 vs 109.3 s) and −49 % in the event surge (46.6 vs 90.6 s); adaptive controllers within 1-2 s of each other in calm conditions
Signals vs routing	cooperative routing alone −24 % under actuated signals (87.5 → 66.9 s); Q-learning vs actuated signals with the same routing −3 % in rush hour (p = 0.08, not significant)
Fair baseline	with the actuated controller tuned on validation runs and the same routing: event surge 46.6 (Q) vs 46.9 s, a tie; rush hour 64.8 vs 57.7 s, tuned actuated better. RL reaches tuned-expert level without manual tuning
AV speed control and lanes	lane selection: AV delay −2.2 %; speed advisory: fewer stops, conflicts −1.1 %, for +6.8 % AV delay
Bus loads	rush-hour buses run full under every holding strategy; a bus every 3 min cuts left-behind passengers from ≈35 to ≈1
High-demand safety	pedestrian max-wait guard at 1.6 × demand: conflicts 18.6 → 10.7 per 1,000 veh (actuated 10.6), delay 179 vs 205 s
Uncertainty	section 3.6: sensing bias (verdict kept in 86 %), VOI ≈ 0, safety-model sensitivity 90 %, re-training 72.5 ± 2.1 s

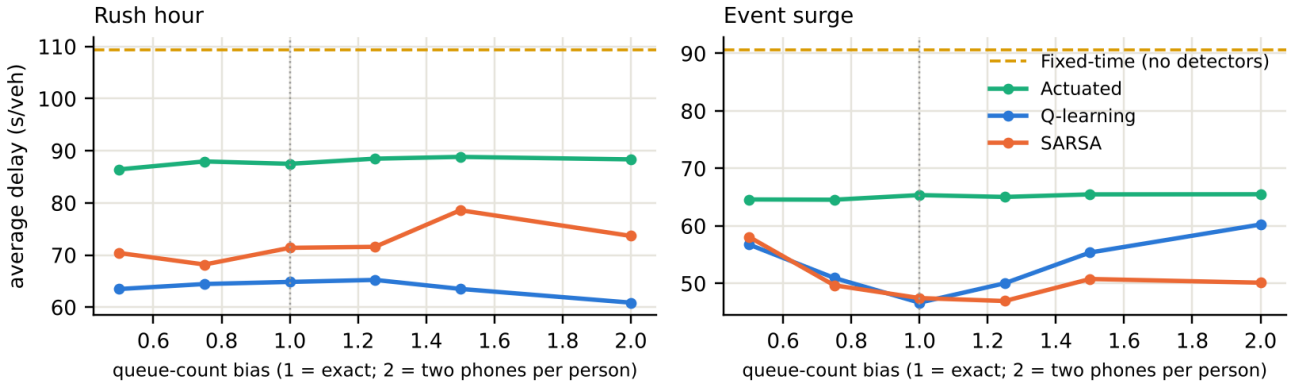


Figure 3. Delay when the signal controllers see biased queue counts (no counting noise). Fixed-time uses no detectors and is unaffected.

Limitations. Synthetic grid not calibrated to observed Dubai counts; 25 % AV share; expected conflicts are a crash surrogate (rankings robust, absolute values not crash rates); agents were trained on exact counts; value of time is a U.S. figure; NOx factors are order-of-magnitude; tabular RL limits state detail. **Next:** calibrate with RTA detector data, train on noisy counts, learn the speed advisory, test a larger network.

Deliverables. This specification; data CSV (data/smart_traffic_simulation_log.csv plus supporting files); Colab notebook (link above); hosted dashboard (link above); presentation (presentation/Smart_Traffic_Group2.pptx) and report; GenAI prompt log (GenAI_disclosure_and_prompts.md).

Reference. Wairagade, A., Ugale, K. & Waghmare, R. (2025). AI-Powered Smart Traffic Management System for Urban Congestion Reduction. *International Journal of Scientific Research & Engineering Trends*, 11(2), 1997-2001. Further sources: Kochenderfer (2015), *Decision Making Under Uncertainty*, MIT Press; Sutton & Barto (2018), *Reinforcement Learning*, 2nd ed., MIT Press; and those in section 2.